



Hands-on Exercise No. 4
DigiSkills 3.0 Batch-03
AI Using Python

Total Marks: 10
Due Date: 18/06/2026

Instructions:

Please read the following instructions carefully before submitting this Hands-on Exercise:

- Use MS Word to prepare exercise solution.
- You may consult tutorials and videos if the concept is not clear.
- Your submitted exercise will not be considered/counted if:
 - It is submitted after due date.
 - It is not in the required format (.doc or .docx)
 - It does not open, or file is corrupt.
 - It is copied (partial or full) from any source (websites, forums, students, etc.)

Learning Outcome:

After completing this exercise, you shall be able to:

- Load, preprocess, and normalize image datasets using **TensorFlow/Keras** for deep learning applications, including normalization, reshaping, and visualization of image samples.
- Design, train, and validate a **Convolutional Neural Network (CNN)** for image classification using multiple convolutional, pooling, and dense layers.
- Evaluate and analyze model performance using accuracy, loss metrics, classification reports, confusion matrices, and prediction visualizations to assess the effectiveness of a Deep Learning model.

Problem Statement

In this exercise, you are required to build, train, and evaluate a **Convolutional Neural Network (CNN)** for image classification using the **Fashion-MNIST dataset**. You will load and preprocess image data, design a multi-layer CNN architecture using TensorFlow/Keras, and train the model to recognize different object categories. After training, you will evaluate the model's performance on unseen test data and visualize the training loss across epochs. The objective of this exercise is to provide hands-on experience with **Deep Learning and Computer Vision**, enabling you to understand the complete end-to-end workflow of an image classification project.

You must use **TensorFlow/Keras only**. **PyTorch is not allowed**. You can use any programming environment such as **Jupyter Notebook, Google Colab, VS Code, or any other Python editor** to write and execute your code. After completing each task, take **screenshots of your code and its output**, and paste them into a **single MS Word (.doc/.docx) file** for submission.

Tasks:

1. Data Loading and Preprocessing: **(Marks 2)**
 - Load the Fashion-MNIST dataset from keras datasets
 - Load the training and test images along with their labels
 - Normalize the pixel values to the range [0, 1]
 - Reshape the images appropriately for input to a CNN (e.g., (28, 28, 1) for grayscale images)
 - Provide a few sample images with their corresponding labels.
2. Design / Build the CNN Model: **(Marks 2)**
 - Design a multi-layer Convolutional Neural Network using TensorFlow/Keras
 - Include at least:
 - 2 convolutional layers
 - MaxPooling layers
 - Dropout for regularization
 - Flatten layer before the dense layers
 - Dense layers leading to the output layer
 - Compile the model using an appropriate optimizer, loss function, and evaluation metric for classification
3. Compile and Train your Model: **(Marks 2)**
 - Train the CNN using the training dataset
 - Use a validation split (e.g., 10–20%) to monitor the model performance during training
 - Train the model for a sufficient number of epochs to achieve stable accuracy
 - Display the training and validation accuracy/loss per epoch

4. Model Evaluation:

(Marks 2)

- Evaluate the trained model on the unseen test dataset
- Report the test accuracy and loss
- Generate a confusion matrix and classification report showing performance on different fashion categories

5. Visualization and Analysis of the results:

(Marks 2)

- Plot the training loss and accuracy curves across epochs
- Plot a few test images along with the predicted and true labels

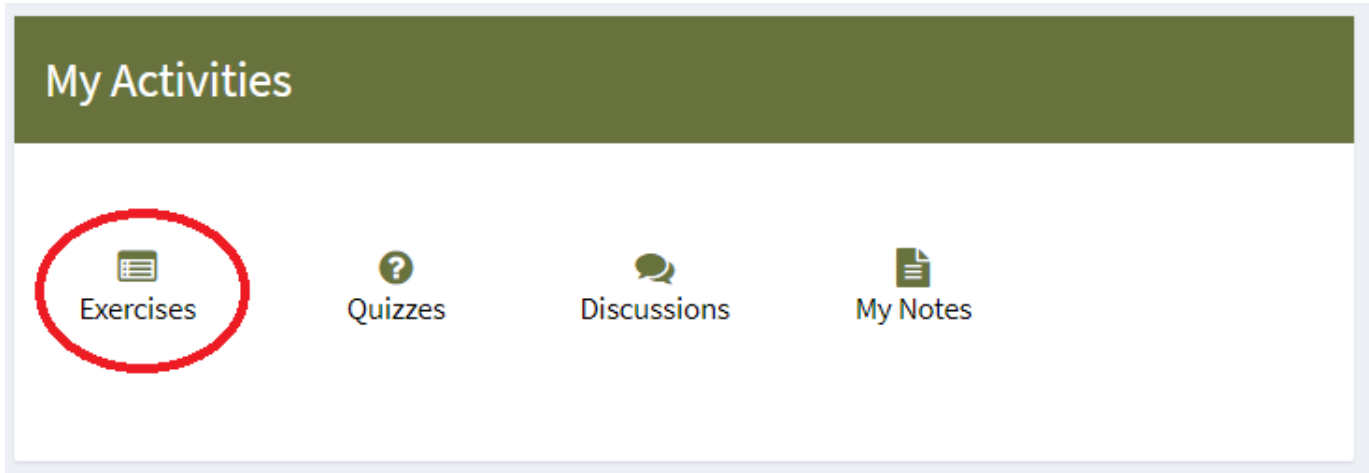
NOTE: In the solution file, no need to send/attach code file; only provide screenshots of the code cells along with their output of above tasks in MS Word File.

BEST OF LUCK 😊

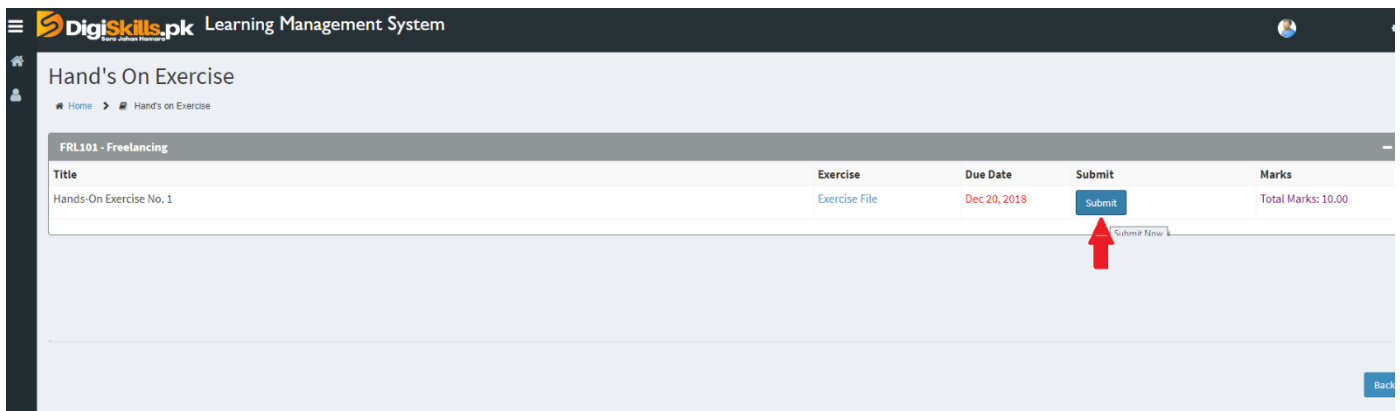
How to submit solution file on LMS?

Please perform the following steps for submitting your solution using LMS:

- 1) Login to the LMS
- 2) Click on the **Exercises** button within the **My Activities** section



- 3) Click on the submit button to upload your Solution.



- 4) Keep in mind to upload your Solution in .doc or .docx format